# Concurrency

Paolo Burgio
paolo.burgio@unimore.it

UNIMORE
UNIVERSITÀ DEGLI STUDI DI
MODENA E REGGIO EMILIA

High Performance
Real Time Lab

# FRATELLI

## QUANDO LA MAMMA ESCE DI CASA

## QUANDO TORNA A CASA

# Why concurrency?

## Functional

- Many users may be connected to the same system at the same time
- Each user can have its own processes that execute concurrently with the processes of the other users
- Perform many operations concurrently

## Performance

- Take advantage of blocking time
- While some thread waits for a blocking condition, another thread performs another operation
- On a multi-core machine, independent activities can be carried out on different cores are the same time

# Competitive vs. Cooperative

**Competitive** concurrency

- Different activities compete for the resources
- One activity does not know anything about the other
- The OS must manage the resources so to
  - Avoid conflicts
  - Be fair

**Cooperative** concurrency

- Many activities cooperate to perform an operation
- Every activity knows about the others
- They must synchronize on particular events

# Competitive

Competing activities need to be “protected” from each other

- Separate memory spaces, as with different processes

The **allocation** of the resource and the synchronization must be **centralized**

- Competitive activities request for services to a central manager (the OS or some dedicated process) which allocates the resources in a fair way

Client/Server model

- Communication is usually done through **messages**

More suitable to the **process** model of execution

# Client/server model

A server manages the resource **exclusively**

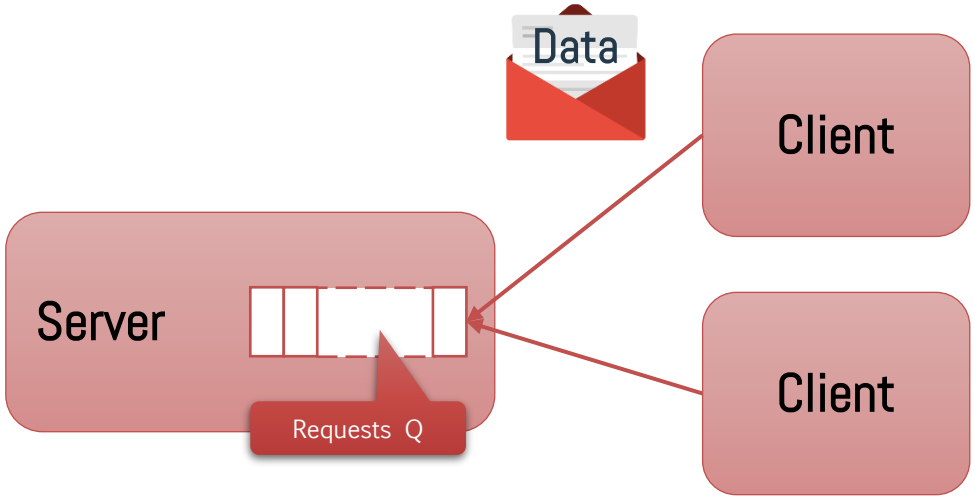
- For example, the printer

If a process needs to access the resource, it sends a **request to the server**

- For example, printing a file, or asking for the status
- The server can send back the responses
- The server can also be on a remote system

Two basic primitives:

- **send** and **receive**



# Cooperative model

Cooperative activities **know** about each other

- They do not need memory protection (less overhead)

They need to access the same data structures

- Allocation of the resource is **de-centralized**
- **Shared memory** model

More suitable to the **thread** model of execution

# Competition vs. cooperation

**Competition** is best resolved by using the **message passing** model

- However, it can be implemented using a shared memory paradigm too

**Cooperation** is best implemented by using the **shared memory** paradigm

- However, it can be realized by using pure message passing mechanisms

General purpose OS needs to support both models

- Protection for competing activities
- Client/server models → message passing primitives
- Shared memory for reducing the overhead

Some special OS supports only one of the two

- RTOS supports only shared memory

# Models of concurrency

# Message passing

Message passing systems are based on the basic concept of message

Two basic operations

```
send (destination, message)
```

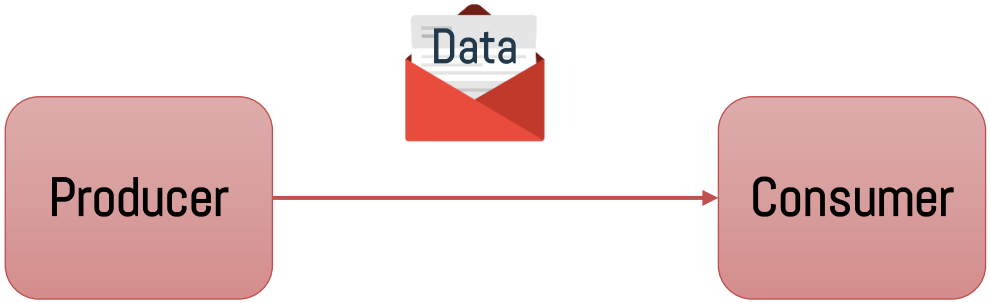
- ✓ send can be synchronous or asynchronous *(fire-and-forget)*

```
receive (source, &message)
```

- ✓ receive can be symmetric or asymmetric

# Producer/Consumer with MP

- ✓ The producer executes `send (consumer, data)`
- ✓ the consumer executes `receive (producer, data)`
- ✓ no need for a special communication structure (already contained in the send/receive semantic)



# Resources and message passing

There are no shared resources in the message passing model

- ✓ all the resources are allocated statically, accessed in a dedicated way

Each resource is handled by a manager process that is the only one that has right to access to a resource

- ✓ The consistency of a data structure is guaranteed by the **manager** process
- ✓ There is no more competition, only cooperation!!!

# Synchronous communication

Synchronous send/receive

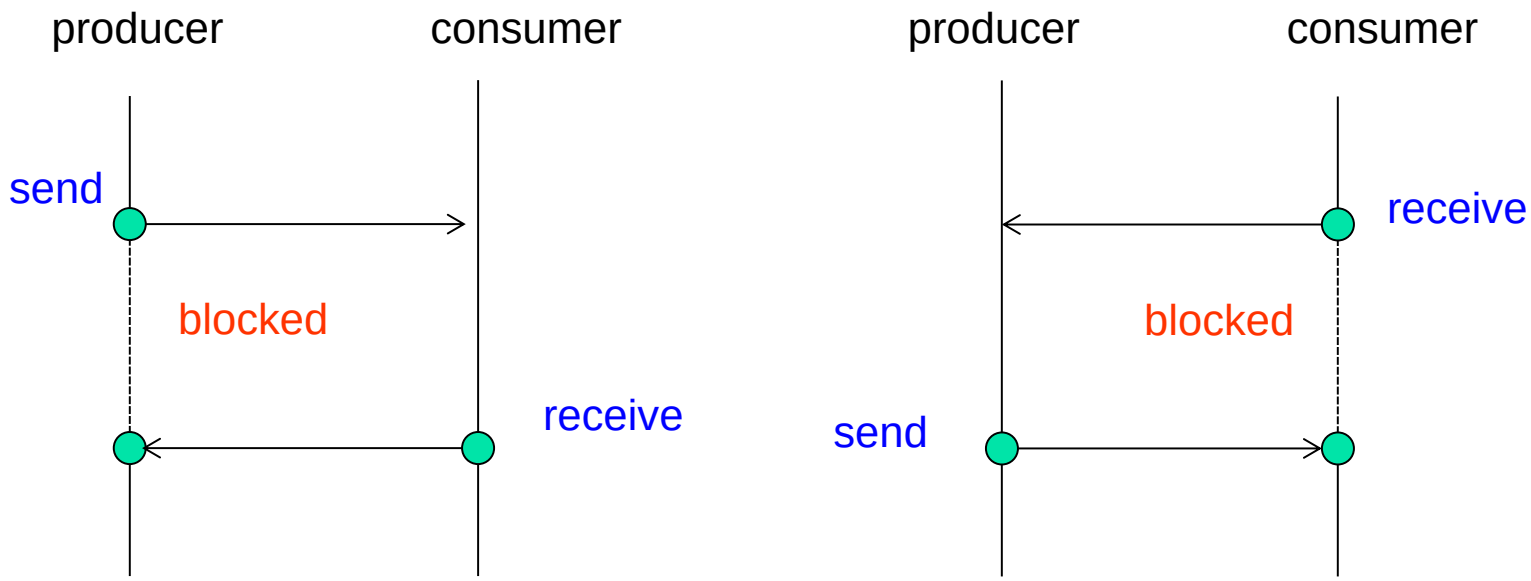
✓ no buffers!

producer:

```
s_send (consumer, d);
```

consumer:

```
s_receive (producer, &d);
```



# Async send/ sync receive

Asynchronous send / synchronous receive

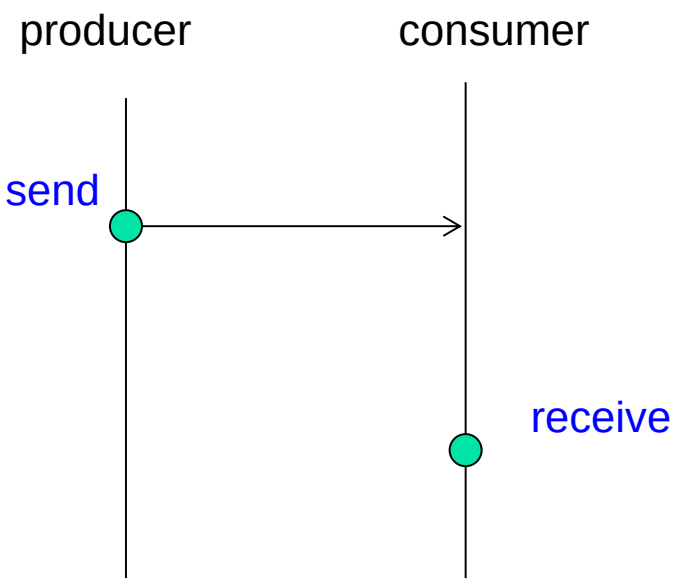
✓ there is probably a send buffer somewhere

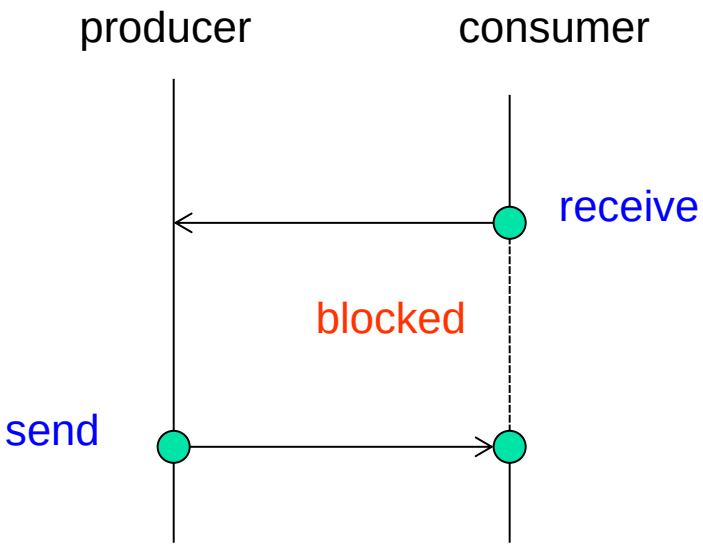
producer:

```
a_send (consumer, d);
```

consumer:

```
s_receive (producer, &d);
```



# (A)symmetric receive

**Symmetric** receive `receive (source, &data);`

✓ the programmer wants a message from a given producer

**Asymmetric** receive `source = receive (&data);`

✓ often, we do not know who is the sender

E.g., a web server is asymmetric

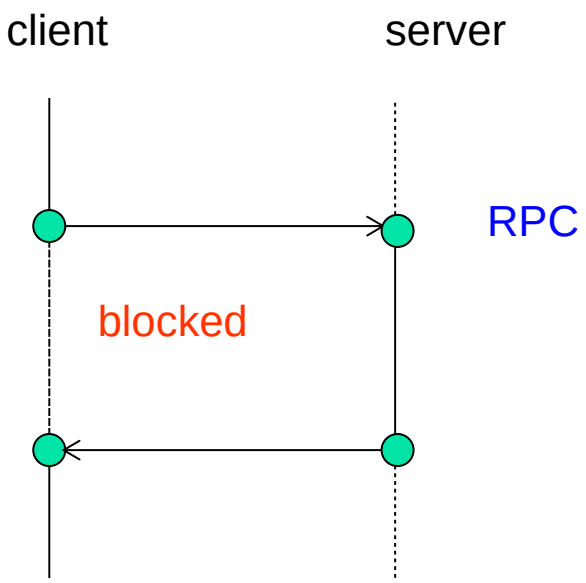
✓ the programmer cannot know in advance the address of the browser that will request the service

✓ many browsers can ask for the same service

# Remote procedure call

From low-level (MP) to more «programmer friendly» (RPC) mechanism

- ✓ Increase expressiveness, higher level of abstraction
- ✓ In a client-server system, a client wants to request an action to a server
- ✓ Typically done using a remote procedure call (RPC)



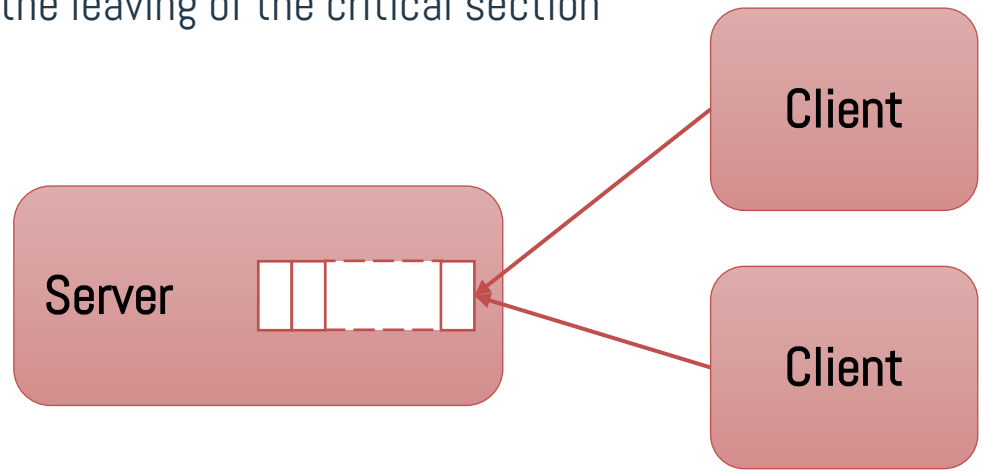
# Massage passing systems

In message passing

- ✓ each resource needs one threads manager (often called **daemon thread**)
- ✓ the threads manager is responsible for giving access to the resource

Example: mutual exclusion with message passing primitives

- ✓ one thread will ensure mutual exclusion
- ✓ Every thread that wants to access the resource must
  - Send a message to the manager thread
  - Access the critical section
  - Send a message to signal the leaving of the critical section



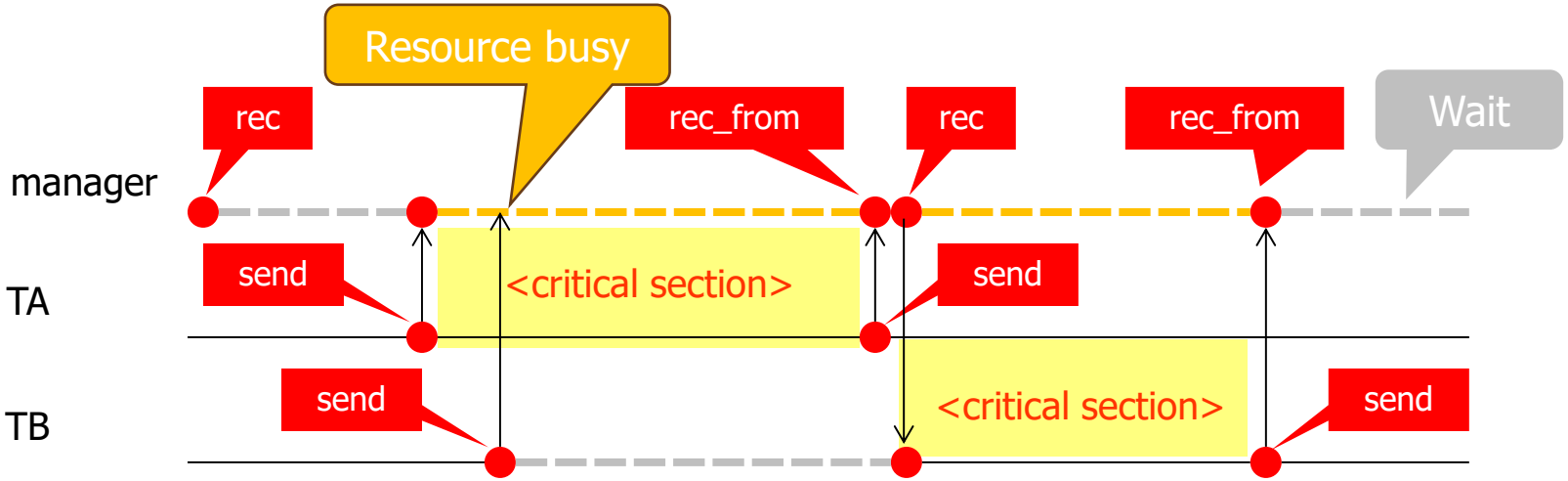
# Sync send / sync receive

```
void * manager(void *) {
  thread_t source;
  int d;
  while (true) {
    source = s_receive (&d);
    s_receive_from (source, &d);
  }
}
```

```
void * thread(void *) {
  int d;
  while (true) {
    s_send(manager, d);

    <critical section>

    s_send(manager, d);
  }
}
```



# With async send and sync receive

```
void * manager(void *) {
  thread_t source;
  int d;
  while (true) {
    source = s_receive (&d);
    a_send (source, &d);
    s_receive_from (source, &d);
  }
}
```

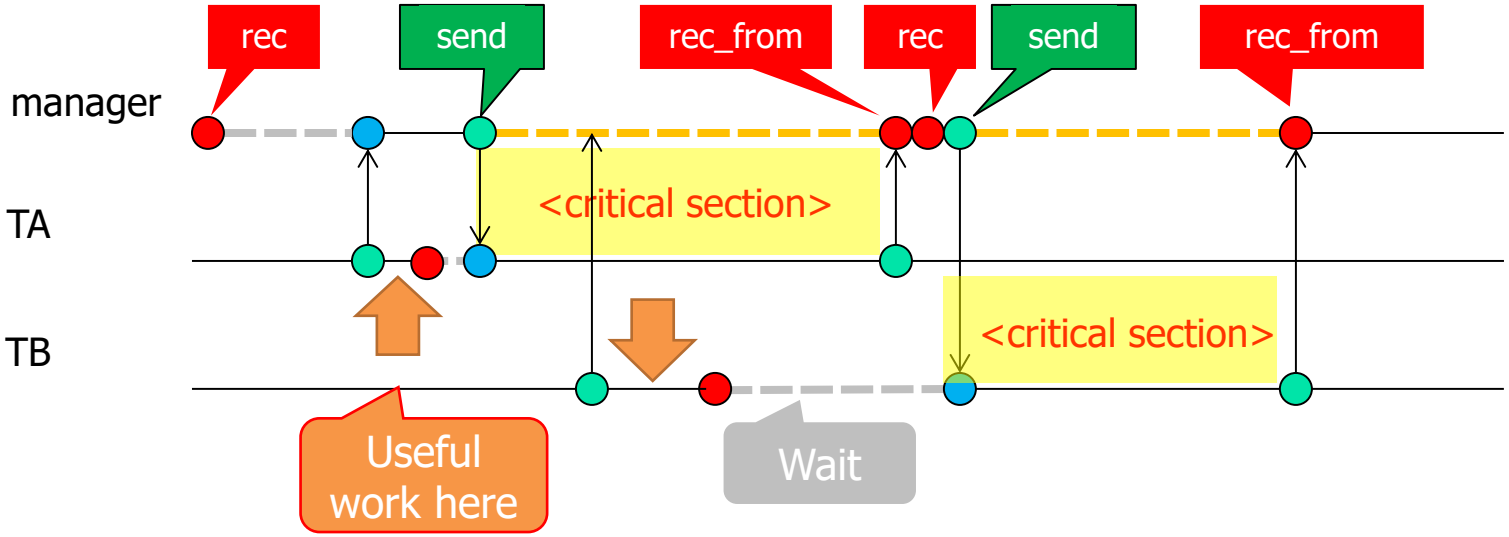
Blocking

```
void * thread(void *) {
  int d;
  while (true) {
    a_send (manager, d);
    s_receive_from (manager, d);

    <critical section>

    a_send (manager, d);
  }
}
```

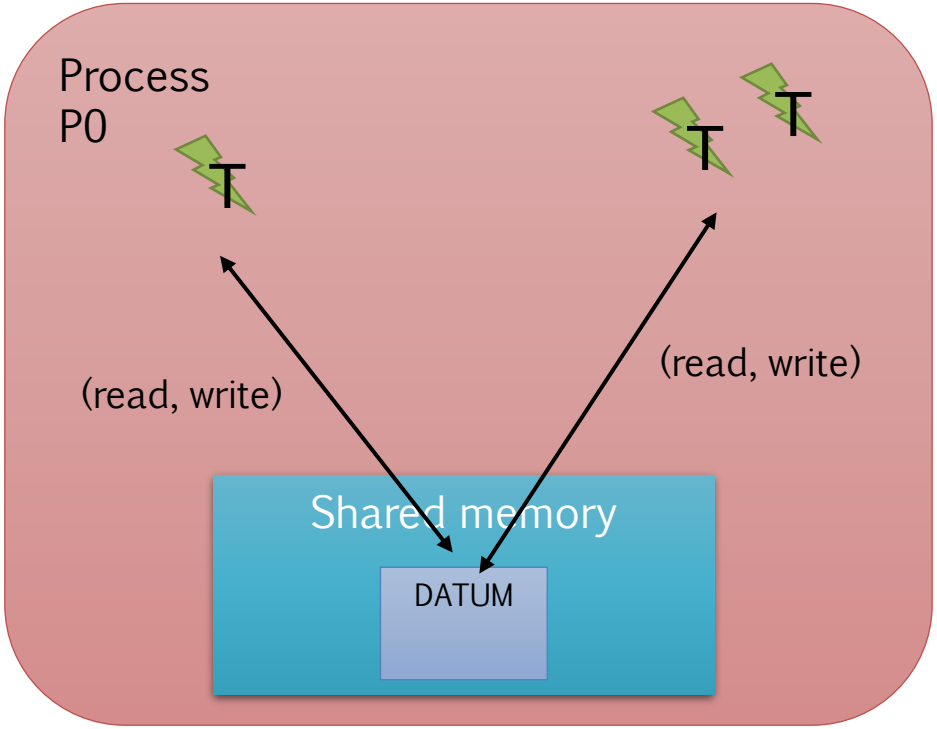
Non blocking



# Shared memory model

# Shared memory abstraction

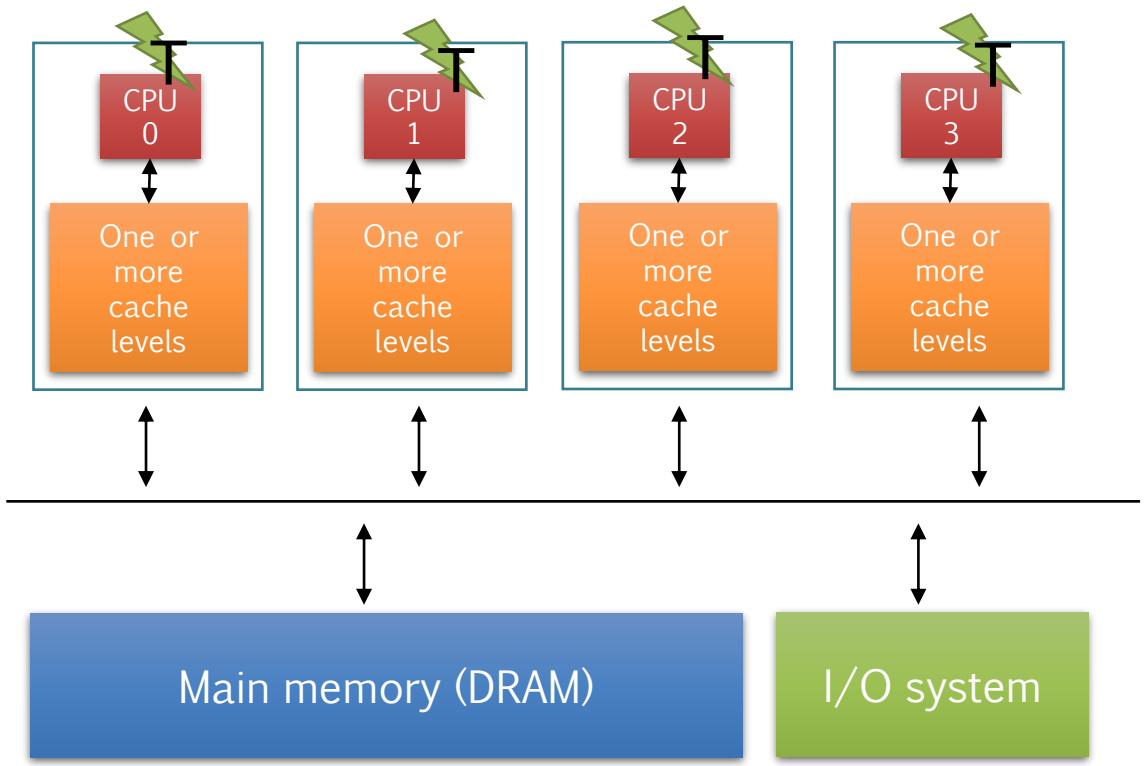
- ✓ The first one being supported in old OS
- ✓ The simplest one and the closest to the machine
- ✓ All threads can access the same memory locations



# Analogy with hardware

An **abstract** model that presents a good analogy is the following:

- Many HW CPU, each one running one activity (thread)
- One shared memory



# Resource allocation

Allocation of resource can be

- ✓ **Static**: once the resource is granted, it is never revoked
- ✓ **Dynamic**: resource can be granted and revoked dynamically
  - Manager

Access to a resource can be

- ✓ **Dedicated**: only one activity at a time may request access to the resource
- ✓ **Shared**: many activities may access the resource at the same time
  - Mutual exclusion

| | **Dedicated** | **Shared** |
|---|---|---|
| **Static** | Compile Time | Manager |
| **Dynamic** | Manager | Manager |

# Mutual exclusion: a (big) problem

We do not know in advance the relative speed of the processes

- ✓ Hence, we do not know the order of execution of the hardware instructions

Example:

- ✓ Incrementing a variable $x$ is NOT an <u>atomic</u> operation

# Atomicity

A hardware instruction is atomic if it cannot be "interleaved" with other instructions

- ✓ Atomic operations are always sequentialized

Atomic operations cannot be interrupted

- ✓ They are <u>thread safe</u> operations
  - For example, transferring one word from memory to register or viceversa

Non atomic operations can be interrupted

- ✓ They are not "safe" operations
- ✓ Non-elementary operations are not atomic

# The big problem

- a++ is not an unique instruction

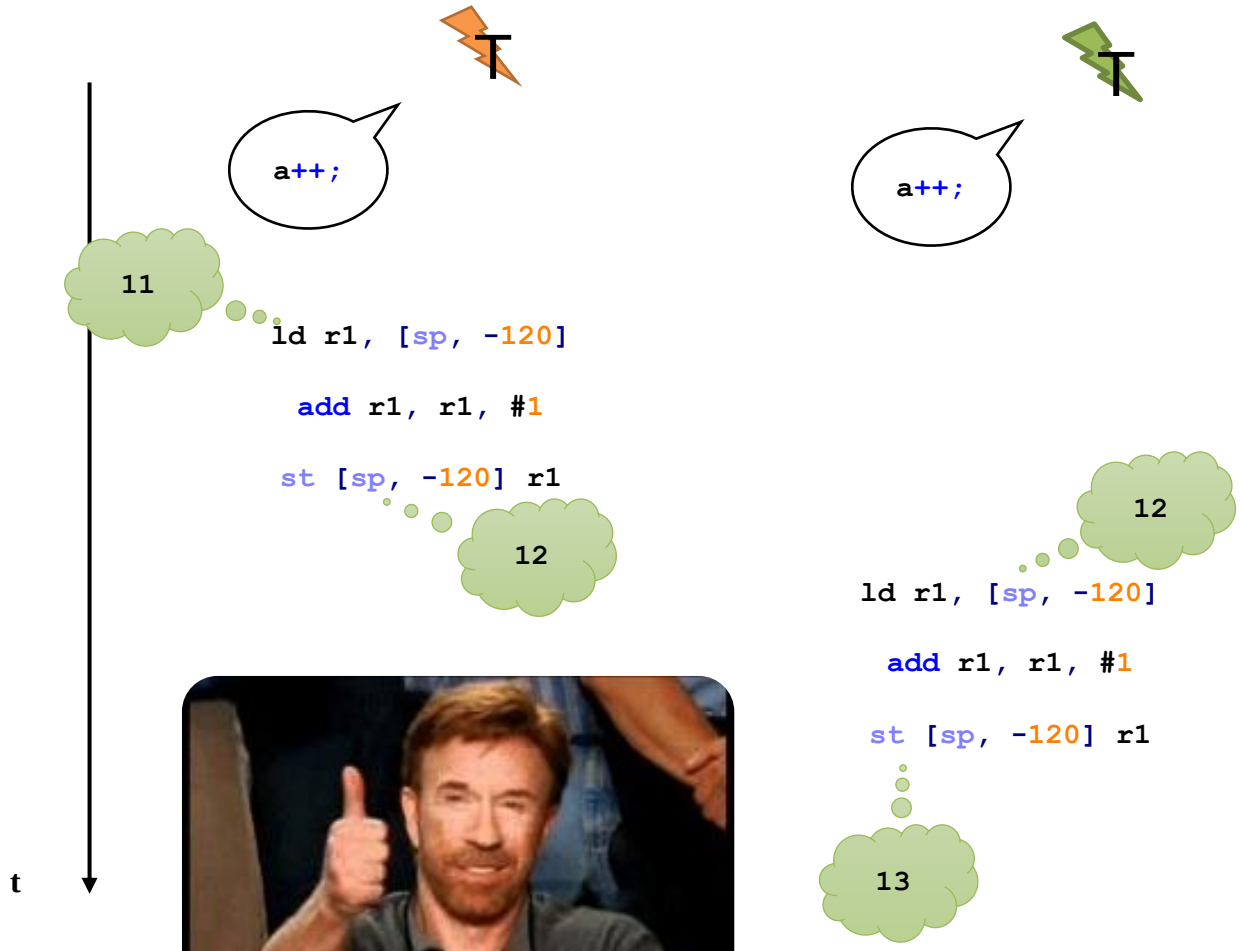
**src.cpp**

```
a++;
```

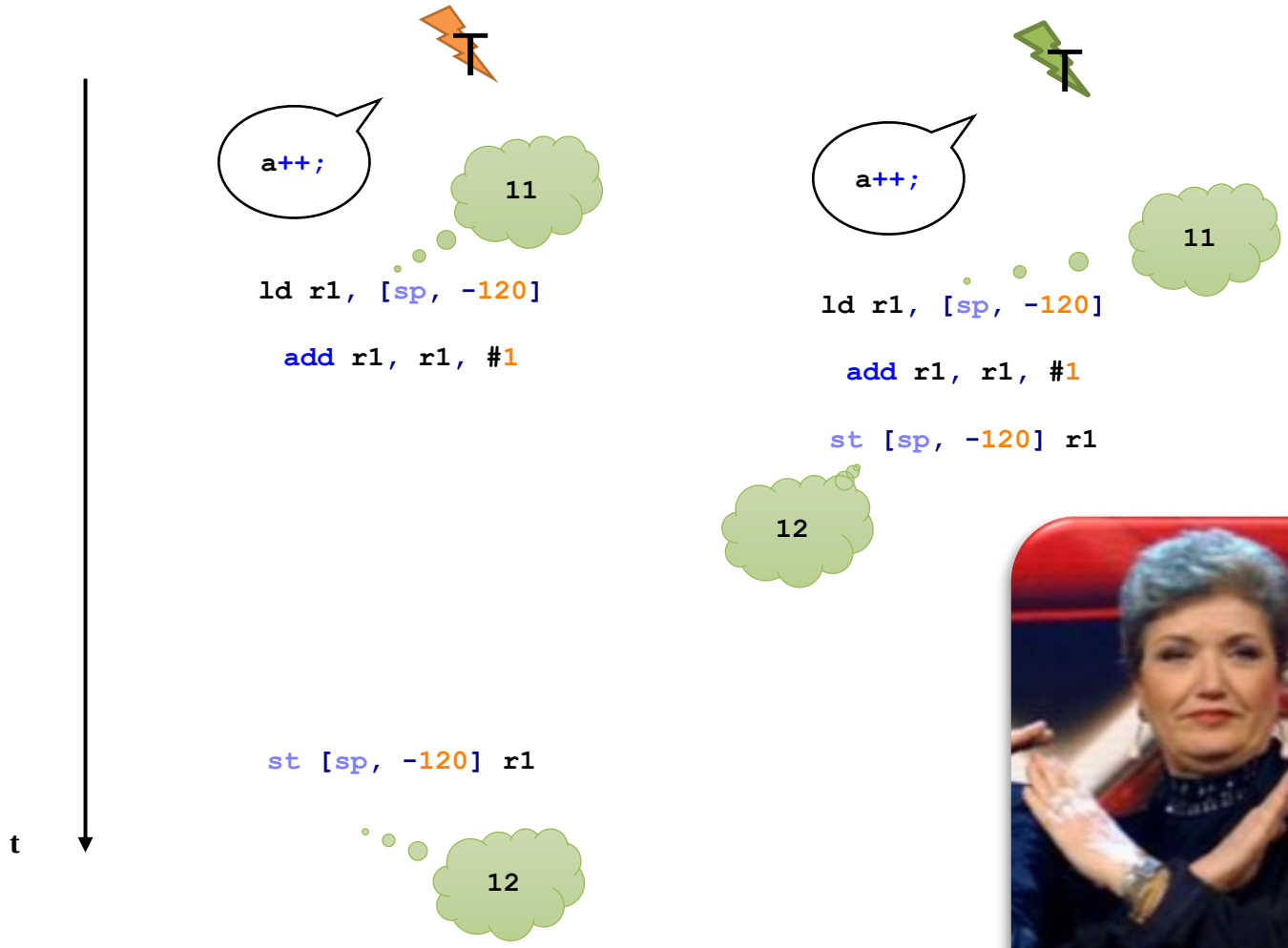
**obj.o**

```
ld r1, [sp, -120] ; load from address of 'a'
add r2, r1, #1 ; add 1
st [sp, -120], r2 ; store in a
```

# When things go well



# When things go less well



# Critical sections

Definitions

- ✓ The shared object (e.g., *x*) where the conflict may happen is a "**resource**"
- ✓ The parts of the code where the problem may happen are called "**critical sections**"

A critical section is a sequence of operations that cannot be interleaved with other operations on the same resource

Multiple critical sections on the same resource must execute in **MUTUAL EXCLUSION**

- ✓ atomic operation (hardware extension within the memory banks)
- ✓ semaphores
- ✓ mutexes
- ✓ *locks*

# General mechanism: semaphores

Proposed by Djikstra

A semaphore is an abstract entity that consists of

- ✓ A counter
- ✓ A blocking queue (of threads)

Can perform two atomic operations

- ✓ Blocking **Wait** for a given condition
- ✓ **Signal** that the condition becomes true (aka **Post**)

We can also use them to implement mutual exclusion (we'll see this)

# Wait and Signal

A **Wait** operation has the following behavior

- ✓ If counter == 0, the requiring thread is blocked
  - It is removed from the ready queue
  - It is inserted in the blocked queue
- ✓ If counter > 0, then counter--;

A **Signal (aka: Post)** operation has the following behavior

- ✓ If counter == 0 and there is some blocked thread, unblock it
  - The thread is removed from the blocked queue
  - It is inserted in the ready queue
- ✓ Otherwise, increment counter

# Semaphores (pseudo-code)

```
void sem_init (sem_t *s, int n) {
  s->count=n;
  // ...
}

void sem_wait(sem_t *s) {
  if (s->count == 0)
    <block the thread>
  else
    s->count--;
}

void sem_post(sem_t *s) {
  if (<there are blocked threads>)
    <unblock a thread>
  else
    s->count++;
}
```

› We assume that `s->count` (and its operation ++) are atomic

# Signal semantics

What happens when a thread blocks on a semaphore?

✓ In general, it is inserted in a BLOCKED queue

Extraction from the blocking queue can follow different semantics:

- ✓ Strong semaphore
  - The threads are removed in well-specified order
  - For example, FIFO order, priority based ordering, …
- ✓ Signal and suspend
  - After the new thread has been unblocked, a thread switch happens
- ✓ Signal and continue
  - After the new thread has been unblocked, the thread that executed the signal continues to execute

Concurrent programs should not rely too much on the semaphore semantic

# Mutual exclusion with semaphores

How to use a semaphore for critical sections?

- ✓ Define a semaphore **initialized to 1**
- ✓ Before entering the critical section, perform a **wait**
- ✓ After leaving the critical section, perform a **signal/post**

```
sem_t s;

// ...
sem_init (&s, 1);
```

```
void *threadA(void *arg) {

  sem_wait (&s);

  <critical section>

  sem_post(&s);
}
```

```
void *threadB(void *arg) {

  sem_wait (&s);

  <critical section>

  sem_post (&s);
}
```

# Mutual exclusion with semaphores

semaphore

counter 0

(TA) sem_wait();
(TA) <critical section (1)>

sem_wait() (TB)

(TA) <critical section (2)>
(TA) sem_post()

<critical section> (TB)
sem_post() (TB)

$t$

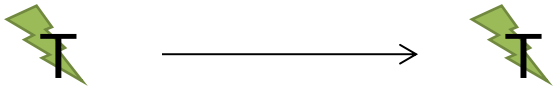
# Synchronisation

Mutual exclusion is not the only problem: we need a way of synchronise two or more threads

- ✓ Example: producer/consumer

We have two threads,

- ✓ One produces some integers and sends them to another thread (**PRODUCER**)
- ✓ Another one takes the integer and elaborates it (**CONSUMER**)



# Synchronization with semaphores

Define a semaphore initialized to 0 (**blocked**)

- ✓ At the synchronization point, consumer performs a **Wait**
- ✓ At the synchronization point, producer performs a **Signal/Post**
- ✓ In the example, *threadA* blocks until *threadB* wakes it up

```
sem_t s;

// ...
sem_init (&s, 0);
```

```
void *threadA(void *arg) {

  sem_wait (&s);

  process (a);
}
```

*<<unlock>>*

```
void *threadB(void *arg) {

  a = 11;

  sem_post (&s);
}
```

# Producer/consumer: how to do it naively

Share a queue of data/objects/anything you might need to produce&consume

- ✓ If the queue is full, the producer actively waits
- ✓ If the queue is empty, the consumer actively waits
- ✓ Aka: **busy-waiting**

Very inefficient!

```
struct CircularArray_t queue;
```

```
void *producer(void *) {
  bool res;
  int data;
  while(1) {
    <obtain data>
    while (!insert_CA(&queue,
                      data))
      ;
  }
}
```

```
void *consumer(void *) {
  bool res;
  int data;
  while(1) {
    while (!extract_CA(&queue,
                       &data))
      ;
    <use data>
  }
}
```

# Naive, polling-based producer/consumer

Consider a producer/consumer system

Producer(s) execute `insert_CA()`

- ✓ We want the producers to be blocked when the queue is full
- ✓ The producers will be unblocked when there is some space again

Consumer(s) execute `extract_CA()`

- ✓ We want the consumers to be blocked when the queue is empty
- ✓ The consumers will be unblocked when there is some space again
- ✓ First attempt: one producer and one consumer only

# One producer, one consumer

```
struct CircularArray_t {
  int array[10];
  int head, tail;
  sem_t empty, full;
}

void init_CA (struct CircularArray_t *c) {
  c->head=0; c->tail=0;
  sem_init(&c->empty, 0); sem_init(&c->full, 10);
}

void insert_CA (struct CircularArray_t *c, int elem) {
  sem_wait(&c->full);
  c->array[c->head] = elem;
  c->head = (c->head + 1) % 10;
  sem_post(&c->empty);
}

void extract_CA (struct CircularArray_t *c, int &elem) {
  sem_wait(&c->empty);
  elem = c->array[c->tail];
  c->tail = (c->tail + 1) % 10;
  sem_post(&c->full);
}
```

*Block if queue is full*

*Release those who're waiting for extraction*

*Block if queue is empty*

*Release those who're waiting for insertion*

# Multiple producers/consumers

Combine mutual exclusion and synchronization

- ✓ Semaphore to implement synchronization
- ✓ Semaphore to protect the data structure

# Producers/consumers: does this work?

```
struct CircularArray_t {
  int array[10];
  int head, tail;
  sem_t full, empty;
  sem_t mutex;
}

void init_CA(struct CircularArray_t *c) {
  c->head=0; c->tail=0;
  sem_init (&c->empty, 0); sem_init (&c->full, 10); sem_init (&c->mutex, 1);
}
```

```
void insert_CA (struct CircularArray_t *c,
                int elem) {
  sem_wait (&c->mutex);
  sem_wait (&c->full);
  c->array[c->head]=elem;
  c->head = (c->head+1)%10;
  sem_post (&c->empty);
  sem_post (&c->mutex);
}
```

*Enter critical section*

*Exit critical section*

```
void extract_CA (struct CircularArray_t *c,
                 int *elem) {
  sem_wait (&c->mutex);
  sem_wait (&c->empty);
  *elem = c->array[c->tail];
  c->tail = (c->tail+1)%10;
  sem_post (&c->full);
  sem_post (&c->mutex);
}
```

...of course NOT!

- Why? (red => protects critical section/mutual exclusion; other colors => synchronization - producer/consumer)

# The deadlock explained

- ✓ A thread executes `sem_wait(&c->mutex)` and then blocks on a synchronisation semaphore
- ✓ To be unblocked another thread must enter a critical section guarded by the same mutex semaphore!
- ✓ So, the first thread cannot be unblocked and free the mutex!

The situation cannot be solved, and the two threads will never proceed

As a rule, **never insert a blocking synchronization** inside a critical section!!!

# Producers/consumers: correct solution

```
struct CircularArray_t {
  int array[10];
  int head, tail;
  sem_t full, empty;
  sem_t mutex;
}

void init_CA(struct CircularArray_t *c) {
  c->head=0; c->tail=0;
  sem_init (&c->empty, 0); sem_init (&c->full, 10); sem_init (&c->mutex, 1);
}
```

```
void insert_CA (struct CircularArray_t *c,
                int elem) {
  sem_wait (&c->full);
  sem_wait (&c->mutex);
  c->array[c->head]=elem;
  c->head = (c->head+1)%10;
  sem_post (&c->mutex);
  sem_post (&c->empty);
}
```

```
void extract_CA (struct CircularArray_t *c,
                 int *elem) {
  sem_wait (&c->empty);
  sem_wait (&c->mutex);
  elem = c->array[c->tail];
  c->tail = (c->tail+1)%10;
  sem_post (&c->mutex);
  sem_post (&c->full);
}
```

...we were mixing producer/consumer and synchronizastion

› NEVER NEST MUTEXES

# References

## Course website

- http://hipert.unimore.it/people/paolob/pub/Industrial_Informatics/index.html

## My contacts

- paolo.burgio@unimore.it
- http://hipert.mat.unimore.it/people/paolob/

## Resources

- Giorgio Buttazzo, "Hard Real-Time Computing Systems : Predictable Scheduling Algorithms and Applications". 3rd Edition. 2011. Springer
- "Real-Time Embedded Systems" course by Prof. Bertogna @UNIMORE
- A "small blog"
  - http://www.google.com